Theoretical Exercise: Linear Kalman Filter

Problem 1: Interpreting Covariance Values

Consider a 1D tracking scenario where the state vector $\boldsymbol{x}_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}$ represents the position p in meters and the velocity v in meters per second of a vehicle along a straight road. You are given three different error covariance matrices representing the uncertainty of the state estimate, assuming the errors are zero-mean Gaussian distributed:

$$a)\ P_k = \begin{bmatrix} 100 & 0 \\ 0 & 0.25 \end{bmatrix}, \quad b)\ P_k = \begin{bmatrix} 4 & 3.5 \\ 3.5 & 4 \end{bmatrix}, \quad c)\ P_k = \begin{bmatrix} 4 & -3.5 \\ -3.5 & 4 \end{bmatrix}$$

(a) What are the standard deviations of the position and velocity estimation errors for a? Interpret this physically: what does a 1σ interval mean for this vehicle's location and speed?

(b) Explain the difference between b and c. In the vehicle scenario, what does a positive off-diagonal element (as in b) imply about how the position and velocity errors are correlated? Based on the physical relationship between position and velocity, which of the two covariance matrices (b or c) may be more realistic in a practical tracking situation? Why?

(c) Imagine you are setting up a Kalman filter for a car that is initially parked somewhere in a 50 m long parking lot, but you are not sure exactly where. The car is definitely at rest. Propose a realistic initial covariance matrix P_0 for this scenario and justify your choice of the diagonal and off-diagonal entries.

Problem 2: Full Predict and Update Step

Consider a discrete-time model of an object moving in 1D with constant velocity subject to random acceleration noise. The state is $\boldsymbol{x}_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}$. The sampling time is $T_\mathrm{s} = 1$ s.

(a) Set up the discrete state transition matrix Φ, and the measurement matrix H assuming only position is measured.

(b) Given the process noise covariance $Q = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, measurement noise variance $R = 2$, initial state $\hat{\boldsymbol{x}}_0^{(+)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and initial covariance $P_0^{(+)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.Calculate the prior estimates $\hat{\boldsymbol{x}}_1^{(-)}$ and $P_1^{(-)}$.

(c) A position measurement $y_1 = 2$ is received at $k = 1$. Calculate the Kalman gain K_1, and the posterior estimates $\hat{\boldsymbol{x}}_1^{(+)}$ and $P_1^{(+)}$.

Problem 3: Steady-State and Convergence

For a time-invariant system, the error covariance matrix P_k may converge to a steady-state value P_∞ as $k \to \infty$.

(a) What mathematical equation does P_∞ satisfy in the discrete-time case?

(b) What are the system-theoretic conditions on the state transition matrix Φ, measurement matrix H, and process noise covariance Q to guarantee that P_k converges to a unique positive semi-definite steady-state P_∞, and that the resulting steady-state Kalman filter is asymptotically stable?

(c) Consider a scalar system with state transition $\Phi = 2$, measurement matrix $H = 1$, process noise variance $Q = 1$, and measurement noise variance $R = 1$. Formulate the scalar DARE and compute the positive steady-state covariance P_∞.

Problem 4: Observability and Error Covariance

Consider the following system:

$$\boldsymbol{x}_{k+1} = \begin{bmatrix} 0.5 & 0 \\ 0 & 2 \end{bmatrix} \boldsymbol{x}_k + \boldsymbol{w}_k, \quad y_k = \begin{bmatrix} 1 & 0 \end{bmatrix} \boldsymbol{x}_k + v_k$$

with process noise covariance $Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and measurement noise variance $R = 1$.

(a) Check the observability of the system using the Kalman observability matrix. Which state is unobservable?

(b) Calculate the prior covariance matrix sequence $P_k^{(-)}$ for a few steps conceptually. What happens to the variance of the unobservable state as $k \to \infty$? Why does this happen physically?

(c) Consider a modified system where the state transition matrix is $\Phi = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}$, while H, Q, and R remain the same. Is this system detectable? Repeat the covariance analysis for the unobservable state. What happens to its variance as $k \to \infty$?